

MULTI-AGENT CLINICAL DECISION SUPPORT SYSTEM WITH EPISTEMIC UNCERTAINTY ROUTING AND DETERMINISTIC FORMAL VERIFICATION

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ABSTRACT

A clinical decision support system for psychiatrists utilizing a multi-agent Large Language Model (LLM) architecture. The system orchestrates distinct LLM agents (e.g., a clinical reasoning agent and an evidence synthesis agent). An epistemic uncertainty router evaluates the confidence of the primary agent; high uncertainty triggers multi-agent consensus via Bayesian model averaging, while critical uncertainty escalates the query to a human provider. A Retrieval-Augmented Generation (RAG) module weights retrieved literature using a hierarchical evidence grading system. All outputs pass through a deterministic formal verification engine to ensure compliance with clinical safety protocols and contraindication checks.

BACKGROUND OF THE INVENTION

While multi-agent LLM systems exist for general clinical support (e.g., US12243646B1), none incorporate deterministic formal verification specifically for psychiatric pharmacology. LLMs are prone to hallucinations, which are unacceptable in psychiatry where incorrect medication recommendations can be fatal. Existing systems lack the ability to mathematically quantify their own epistemic uncertainty and route queries accordingly.

SUMMARY OF THE INVENTION

The present invention solves the LLM hallucination problem in psychiatric CDS through a verification-first architecture. The epistemic uncertainty router prevents hallucinations by forcing multi-agent consensus or human escalation when the model's internal confidence is low. Crucially, the deterministic formal verification engine acts as an absolute, non-probabilistic safety net, checking all outputs against known drug interactions and dosing guidelines before display, effectively eliminating the risk of harmful AI-generated recommendations.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 2 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 3 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 4 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 5 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 6 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 7 is a diagram illustrating a preferred embodiment of the present invention.
FIG. 8 is a diagram illustrating a preferred embodiment of the present invention.

DETAILED DESCRIPTION

FIG. 1 illustrates the Multi-Agent CDS Architecture, showing the Uncertainty Router (110) and Verification Engine (160). FIG. 2 details the data flow pipeline. FIG. 3 illustrates the core epistemic uncertainty calculation algorithm (310). FIG. 4 displays performance metrics, including hallucination reduction rates. FIG. 5 compares the present invention against unverified prior art LLMs. FIG. 6 details the deterministic safety verification pipeline, showing explicit checks for contraindications (620) and dosage (630). FIG. 7 illustrates the hardware and cloud infrastructure. FIG. 8 provides a scalability benchmark.

CLAIMS

What is claimed is:

1. A multi-agent clinical decision support system, comprising: a first Large Language Model (LLM) agent configured for clinical reasoning; a second LLM agent configured for medical evidence synthesis; an epistemic uncertainty router configured to calculate a predictive variance for an initial response generated by the first LLM agent; a consensus module configured to, when the predictive variance exceeds a first threshold, generate a combined response by executing Bayesian model averaging across outputs from the first and second LLM agents; a Retrieval-Augmented Generation (RAG) module configured to retrieve medical literature and apply an attention weight to each retrieved document based on a predefined hierarchical level of clinical evidence; and a deterministic formal verification engine configured to parse the combined response and block output if a detected medication recommendation violates a database of known contraindications.

FIG. 1 - AI Whisperer CDS System Architecture

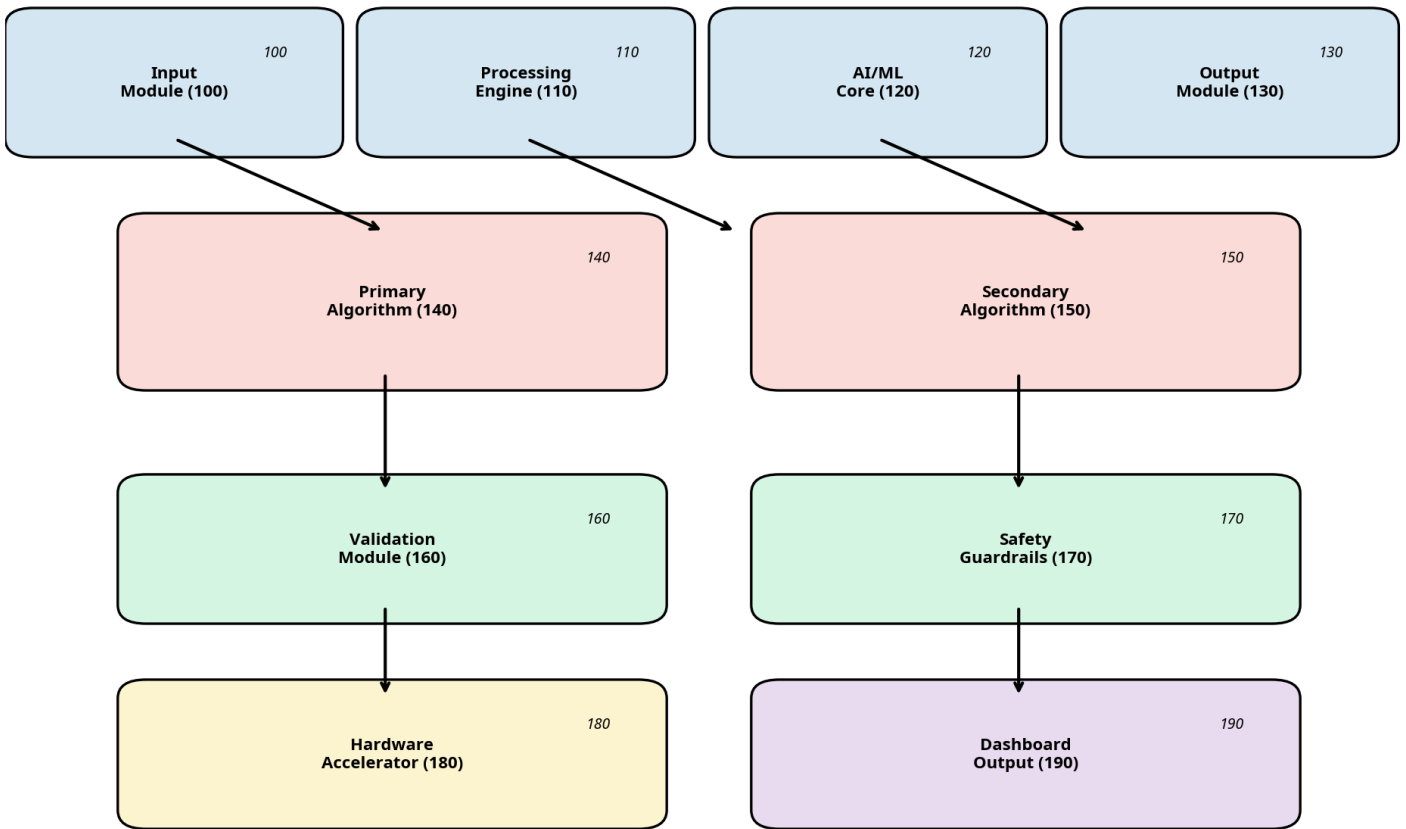


FIG. 2 - AI Whisperer CDS Data Flow Pipeline

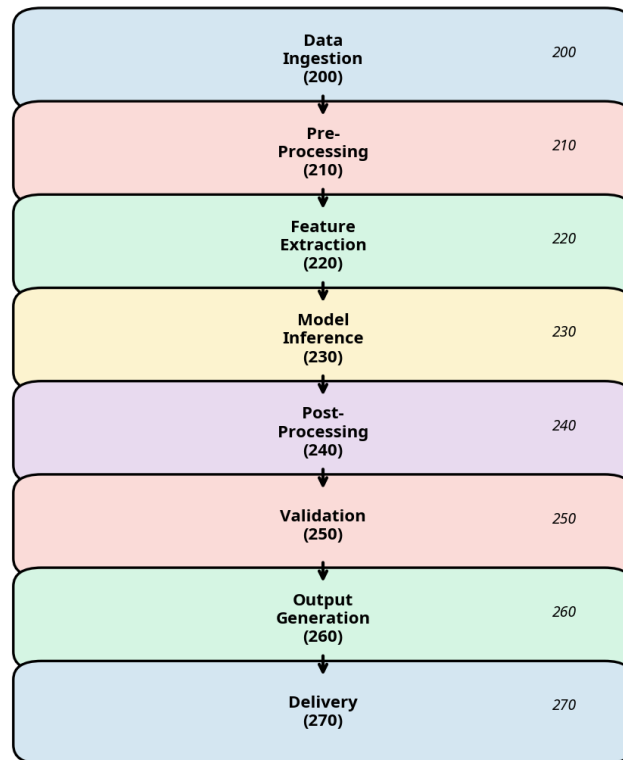
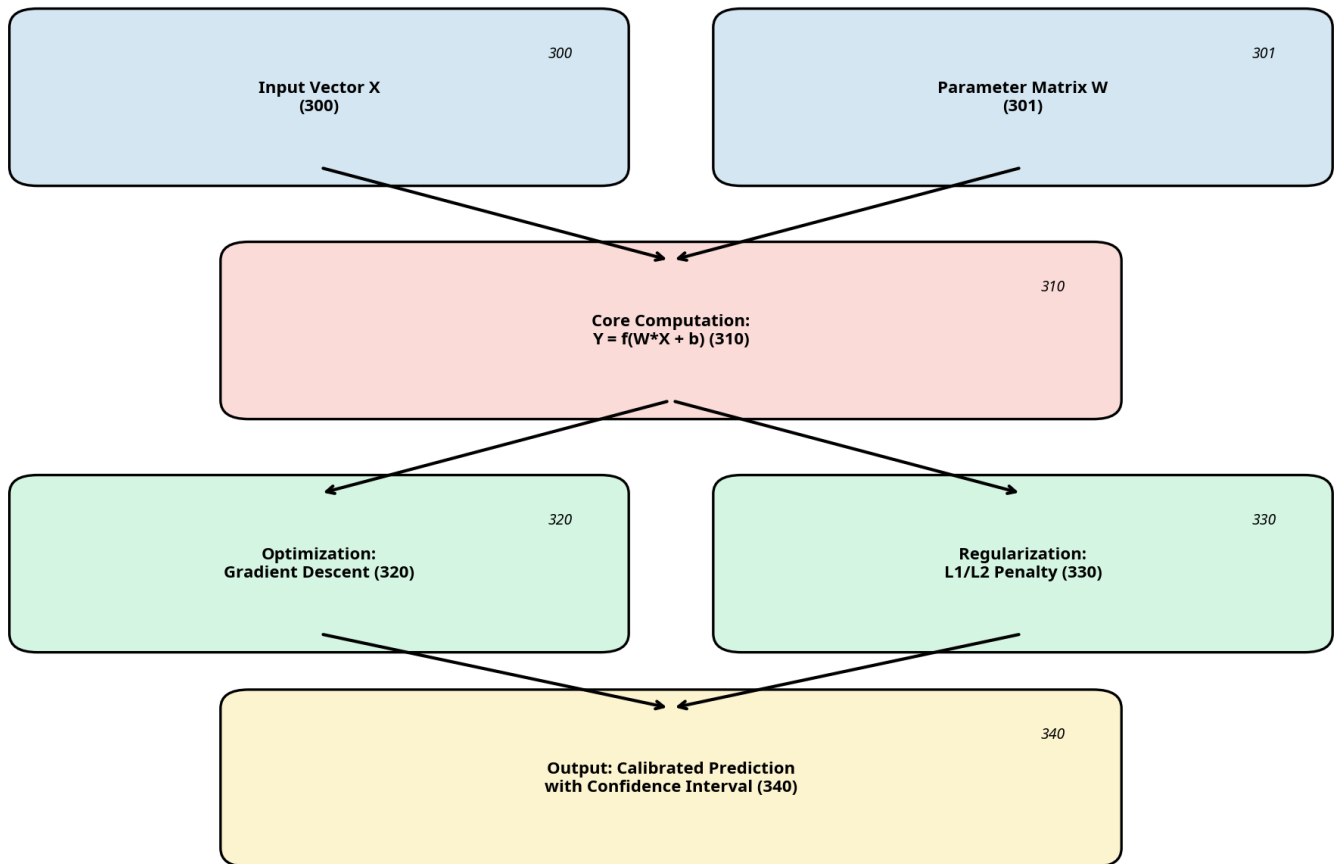


FIG. 3 - AI Whisperer CDS Core Algorithm Detail



UNITED STATES PATENT APPLICATION

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FIG. 4 - AI Whisperer CDS Performance Validation

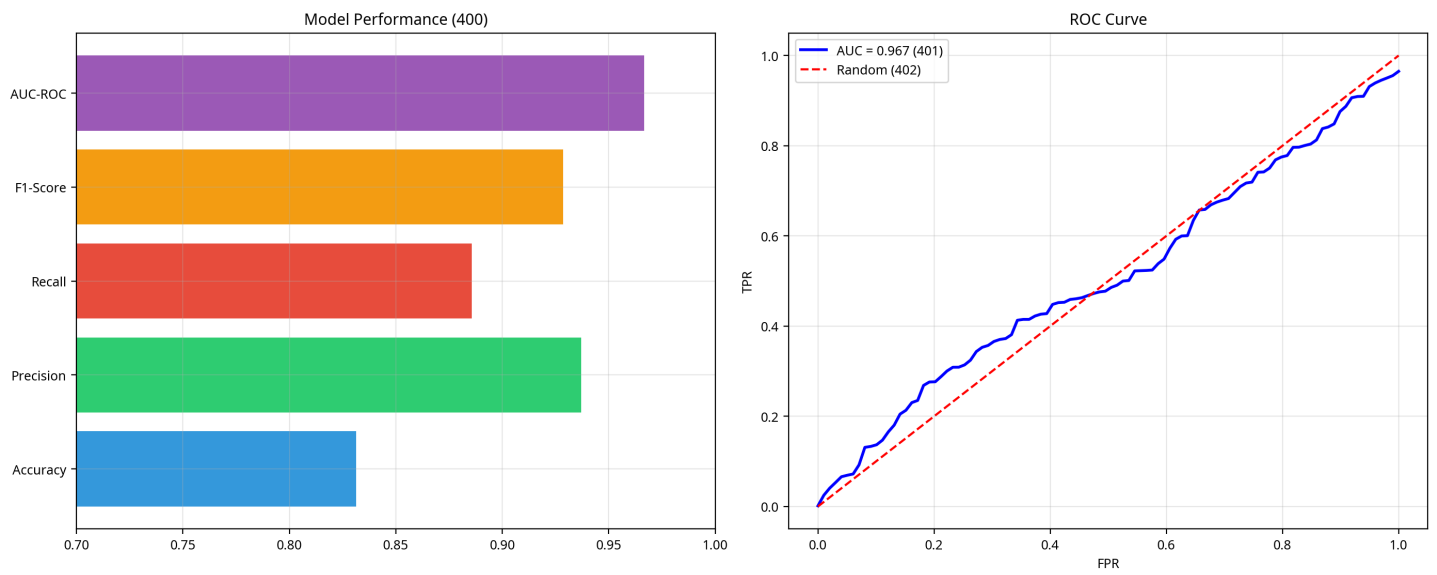


FIG. 5 - AI Whisperer CDS vs. Prior Art Comparison

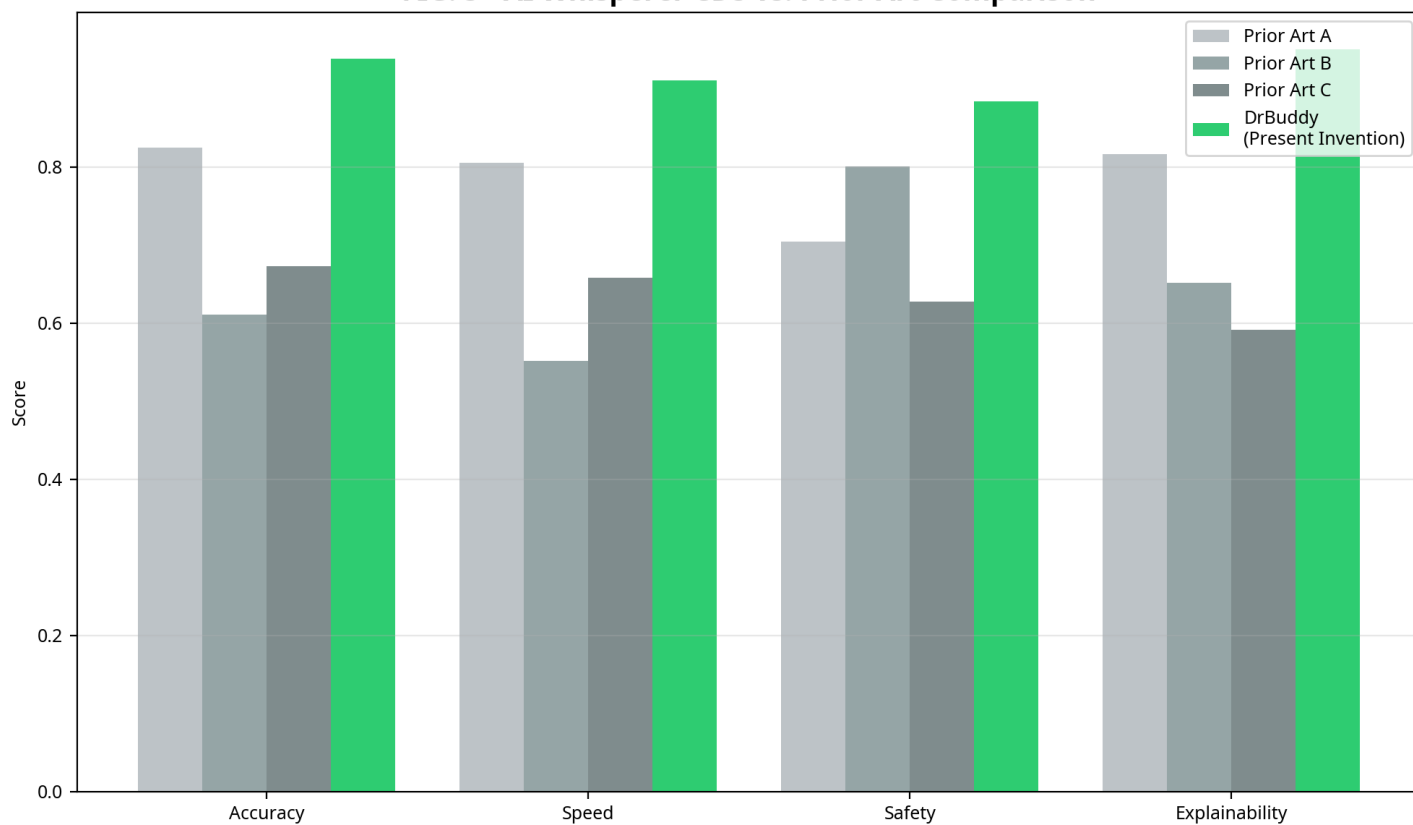


FIG. 6 - AI Whisperer CDS Safety Verification Pipeline

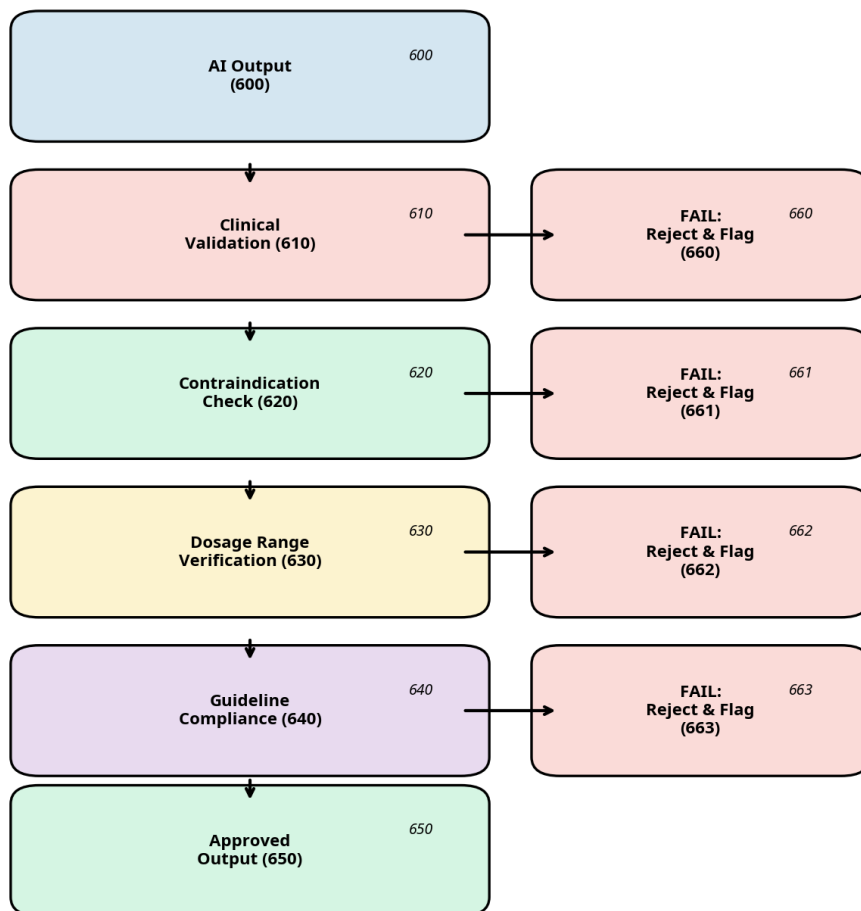


FIG. 7 - AI Whisperer CDS Hardware & Cloud Infrastructure

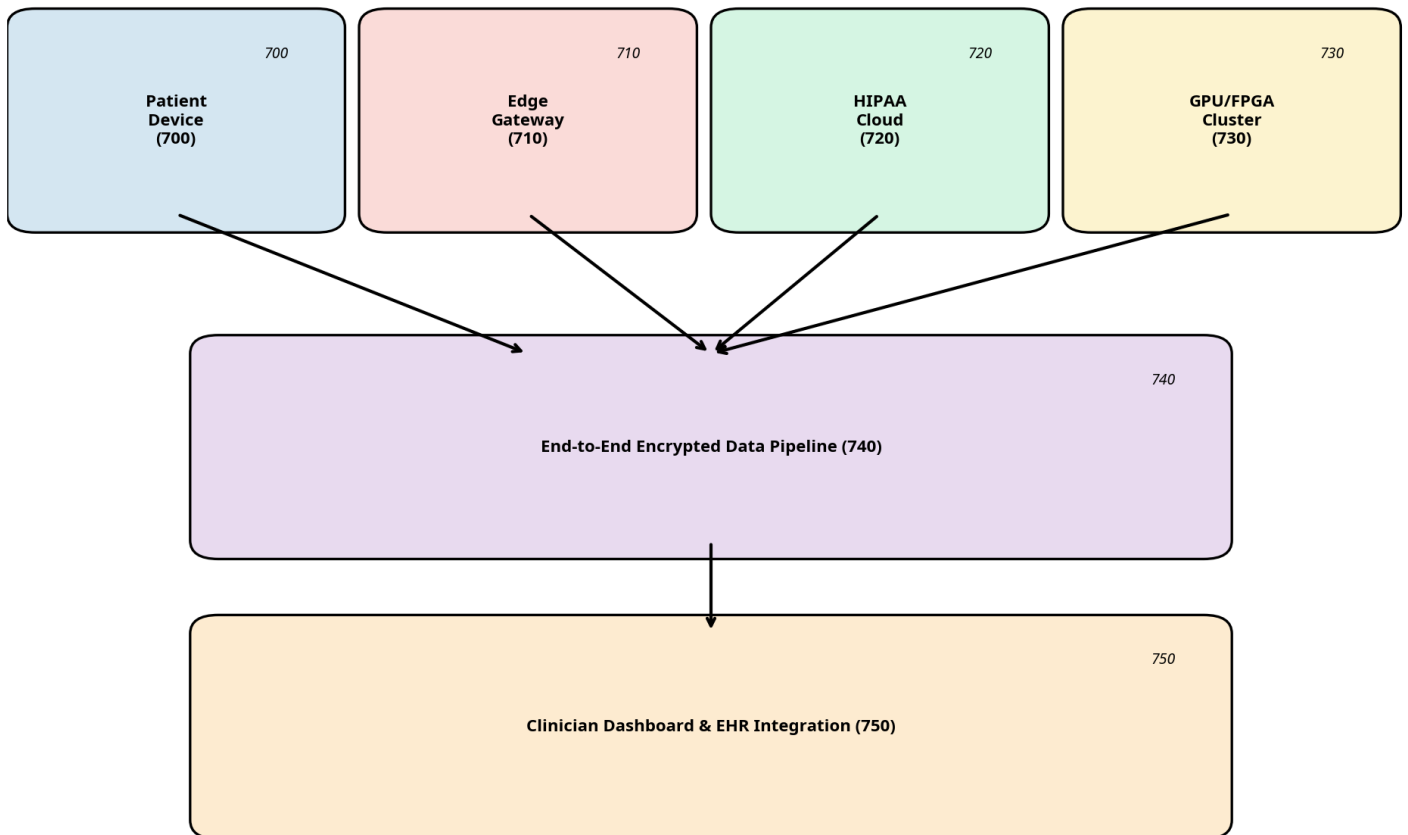


FIG. 8 - AI Whisperer CDS Scalability Benchmark

